

First Problem Sheet

on

Linear Algebra

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Topic: Linear Spaces

“Mathematics is the art of reducing any problem to linear algebra.”
-William Stein

Problem 1

Prove the following statements.

- i) The identity element of a group is unique.
- ii) The inverse of a given element of a group is unique.
- iii) The inverse of a product is the product of the inverses in reversed order.
- iv) The cancellation law holds true: If Y, Z are elements of a given group such that

$$XY = XZ$$

for some element X in the group, then $Y = Z$. Similarly,

$$YX = ZX$$

implies the same conclusion.

- v) For any group G we have $fG = G$ for any $f \in G$ where $fG = \{fg : g \in G\}$.

Problem 2

The definition of a linear space is usually written in terms of an explicit list of rules that a set equipped with binary operations must satisfy. Using our definition of a linear space (in terms of semigroups, groups, and bodies) over a body, state the complete explicit list of axioms that set must satisfy in order for it to be a linear space. Such list is usually called the **linearity axioms**.

Problem 3

Show that bullets ii), iii), v), ix) in Example 1.5 are all indeed examples of linear spaces.

Hint: for bullet ix) you may appeal to the usual theorems for combinations of continuous function from complex analysis.

Problem 4

Let $\mathcal{V}$ be a linear space over $\mathbb{K}$. For $\alpha \in \mathbb{K}$ and $v \in \mathcal{V}$, show that $\alpha v = 0$ implies $\alpha = 0$ or $v = 0$.

Problem 5

Let ∞ and $-\infty$ be two distinct objects, neither of which is in $\mathbb{R}$. Define an addition and scalar multiplication on $\mathbb{R} \cup \{\infty, -\infty\}$ by

$$t\infty = \begin{cases} -\infty & t < 0, \\ 0 & t = 0, \\ \infty & t > 0, \end{cases} \quad t(-\infty) = \begin{cases} \infty & t < 0, \\ 0 & t = 0, \\ -\infty & t > 0, \end{cases}$$

and

$$\begin{aligned} t + \infty &= \infty + t = \infty + \infty = \infty, \\ t + (-\infty) &= (-\infty) + t = (-\infty) + (-\infty) = -\infty, \\ \infty + (-\infty) &= (-\infty) + \infty = 0. \end{aligned}$$

With these operations of addition and scalar multiplication, is $\mathbb{R} \cup \{-\infty, \infty\}$ a linear space over $\mathbb{R}$?

Note: These definitions (except the last one, where $-\infty + \infty = 0$, are actually used in Measure Theory to define the set of extended real numbers and work with the Lebesgue integral).

Problem 6

Let $\beta = \{v_j\}_{j=1}^n$ and let $\beta_m = \{v_j\}_{j=1}^m$ for $m = 1, \dots, n$. Show that β is a linearly independent set if and only if $v_k \notin \text{span } \beta_{k-1}$ for all $k \geq 1$ (here, we define $\beta_0 = \emptyset$ and understand that $\text{span } \emptyset = \{0\}$).

Problem 7

For each of the following subsets of $\mathbb{C}^3$, determine whether it is a subspace of $\mathbb{C}^3$.

- i) $\{(\alpha_1, \alpha_2, \alpha_3) : \alpha_1 + 2\alpha_2 + 3\alpha_3 = 0\}$
- ii) $\{(\alpha_1, \alpha_2, \alpha_3) : \alpha_1 + 2\alpha_2 + 3\alpha_3 = 4\}$
- iii) $\{(\alpha_1, \alpha_2, \alpha_3) : \alpha_1\alpha_2\alpha_3 = 0\}$
- iv) $\{(\alpha_1, \alpha_2, \alpha_3) : \alpha_1 = 5\alpha_3\}$

Moreover, identify (geometrically) the subspaces in i) and iv).

Problem 8

Show that the subset S of $\mathcal{M}^{2 \times 2}$ of all symmetric matrices is a linear subspace of $\mathcal{M}^{2 \times 2}$. Find a basis for S and thus show that this subspace is three-dimensional.

Problem 9

Show that the intersection of two subspaces of a linear space is a subspace of that same linear space.

Problem 10

Prove Theorem 1.8, 1.9, and 1.11.

Problem 11

Given a basis $\{a_i\}_{i=1}^N$ for a linear space $\mathcal{V}$, construct an orthonormal basis by taking linear combinations of the original one. More specifically, find a basis $\{e_i\}_{i=1}^N$ such that $\|e_i\| = 1$ for all i , such that $g(e_i, e_j) = \delta_{ij}$ and $e_i = \alpha_j a_j$ for some set $\{\alpha_j\}_{j=1}^N$ of scalars for each element e_i . This procedure is usually called the **Gram-Schmidt process**.

Problem 12

Show that for any set of n complex numbers $\{\alpha_j\}_{j=1}^n$ we have

$$\left| \sum_{i=1}^n \alpha_i \right|^2 \leq n \sum_{i=1}^n |\alpha_i|^2.$$

Hint: Apply the Schwarz inequality to $(\alpha_1, \alpha_2, \dots, \alpha_n)$ and another appropriate vector.

Problem 13

Verify that $\mathbf{C} \in \text{End}(\mathcal{V})$ where $\mathcal{V}$ is $\mathbb{C}$ over $\mathbb{R}$ and

$$\mathbf{C}|z\rangle = |z^*\rangle.$$

Is $\mathbf{C}$ linear if instead of real numbers, complex numbers are used as scalars?

Problem 14

Define the **kernel** of $\mathbf{T} \in \text{Hom}(\mathcal{V}, \mathcal{W})$ as

$$\ker \mathbf{T} = \{|v\rangle \in \mathcal{V} : \mathbf{T}|v\rangle = |0\rangle\},$$

and hence prove that $\ker \mathbf{T}$ is a subspace of $\mathcal{V}$ and that $\mathbf{T}(\mathcal{V})$ is a subspace of $\mathcal{W}$.

Problem 15

Show that the map $d : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$ defined as in Theorem 1.29 is indeed a metric.

Problem 16

Show that the map $\|\cdot\| : \mathcal{V} \rightarrow \mathbb{R}$ defined as in Theorem 1.31 is indeed a norm.

Problem 17

Show that if $(\alpha_n), (\beta_n)$ are in ℓ^2 , then

$$\sum_{j \in \mathbb{N}} \alpha_j^* \beta_j$$

is convergent.

Problem 18

Express the following permutations as products of disjoint cycles, and determine which are cyclic.

i) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 3 & 4 & 5 & 6 & 2 \end{pmatrix}$

ii) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 4 & 5 & 6 & 3 \end{pmatrix}$

iii) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 5 & 4 & 2 \end{pmatrix}$

Problem 19

Show that if $\pi \in S_n$, then π and π^{-1} have the same parity.

Problem 20

Prove that the determinant of a linear operator $\mathbf{A}$ has the following properties:

- i) If $\mathbf{A} = \lambda \mathbf{1}$, then $\det \mathbf{A} = \lambda^N$.
- ii) $\mathbf{A}$ is invertible if and only if $\det \mathbf{A} \neq 0$.
- iii) $\det(\mathbf{A} \circ \mathbf{B}) = \det \mathbf{A} \det \mathbf{B}$.

Problem 21

Let Δ be a determinant function in the N -dimensional linear space $\mathcal{V}$. Let $|v\rangle$ and $\{|v_k\rangle\}_{k=1}^N$ be vectors in $\mathcal{V}$. Then show that

$$\sum_{j=1}^N (-1)^{j-1} \Delta \left(|v\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, |v_N\rangle \right) |v_j\rangle = \Delta(|v_1\rangle, \dots, |v_N\rangle) |v\rangle$$

where a hat on a vector means that particular vector is missing.

Problem 22

Show that the classical adjoint satisfies the following relations:

$$\text{ad}(\mathbf{A}) \circ \mathbf{A} = \det \mathbf{A} \cdot \mathbf{1} = \mathbf{A} \circ \text{ad}(\mathbf{A}).$$

Where $\mathbf{1}$ is the identity operator.